

Lab 06

(Programming Revision)

Putting together everything learnt so far...

Objectives

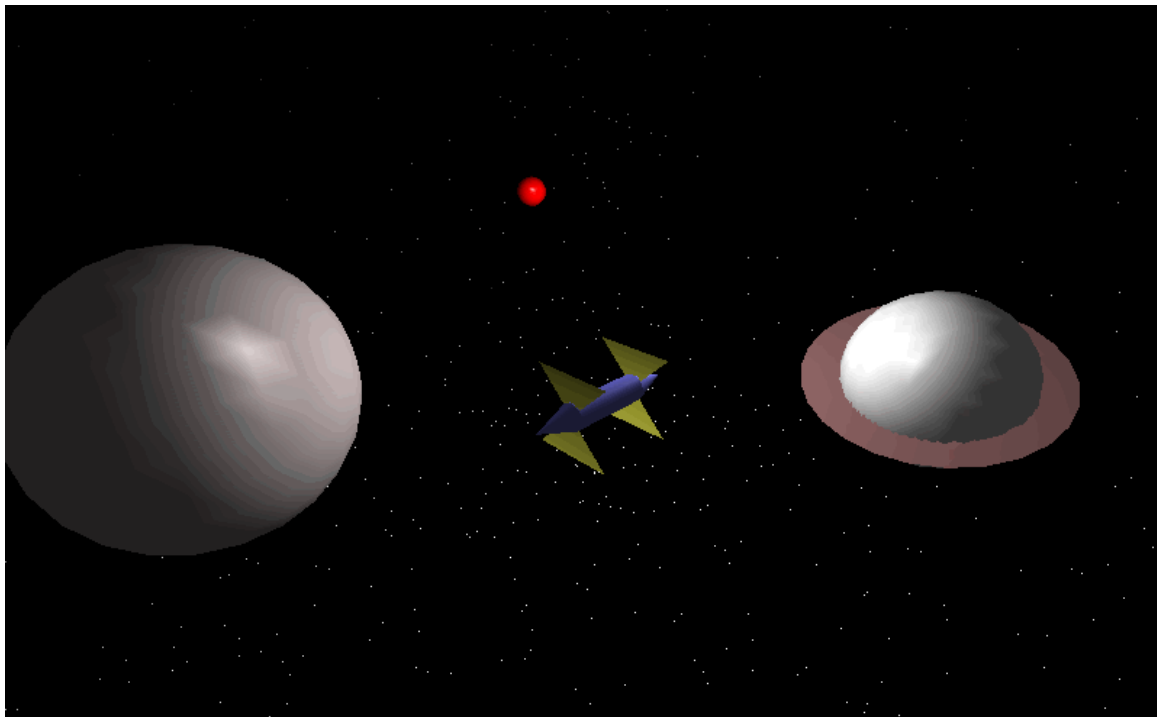
- We will revise on the basic topics that we have covered so far (basic primitives, quadrics and object transformations).
- We will also learn how to create objects in a 3D world based on a given expected output. This tests your observation skill and ability to construct meaningful objects from ground up.

Interim Evaluation

(80 minutes)

For this lab exercise, you will have to create a simple outer space world, consisting of distant stars, some planets and a hovering spaceship.

But this time, source codes are NO LONGER provided for you! You will have to create your 3D world and its objects to be as similar (or better!) as possible to the given screenshot shown below:



Starting off

As usual, you can start off by using the program that you have created in the previous lab exercises.

You should know by now that you need to create the new `CGLab06.hpp` and `CGLab06.cpp` files in order to create new classes and their member functions for modelling and drawing your objects in this lab. If you are not sure, please refer back to the “**Revision regarding creating classes**” section at the beginning of Lab04.

Start off by creating a class with some member functions. A straightforward suggestion is to create a class call **MyPlanets**, and separate the objects you want to create into 3 different member functions (for more modularity and easier separation of code), for e.g. **drawStars()**, **drawPlanets()** and **drawShip()**. This is just a suggestion. You may create and name them in any way you wish.

Construct and Move

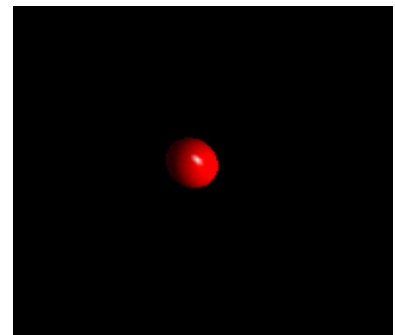
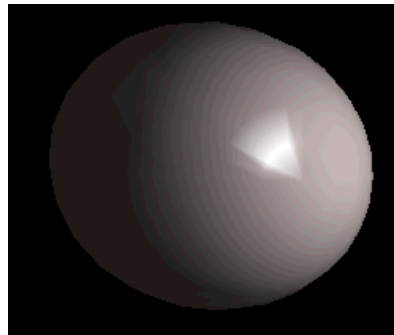
One good tip when you have many objects to create is to first construct them at the origin (of the axis), then only move it to the location that you want using appropriate transformations.

Some Hints to Help You

Begin with the sky. You can create a 3D space filled with random white points. Look back at the earlier tutorials on how this can be done.

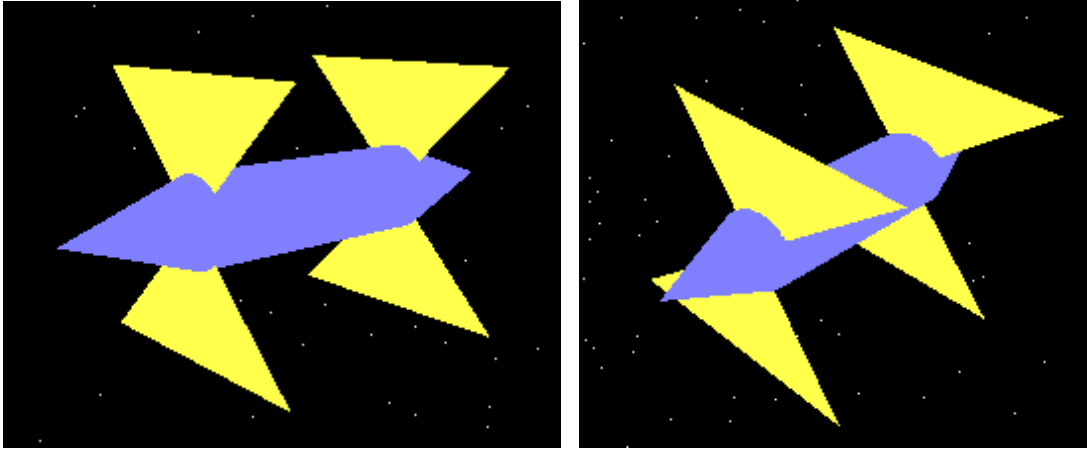


Next, try and construct Saturn, Jupiter and Mars. Jupiter and Mars can be easily created using spheres. Saturn has an additional outer disc. All these planets can be easily constructed using quadrics.



(Shading mode on)

And finally, create a simple spaceship that looks like the one below. You will need a cylinder for the body, 2 cones attached to both ends of the body, and 4 triangles for its wings. You could use some creativity here to make this spaceship look more sophisticated!



(Shading mode off)

Too Easy?

What else can you add to your outer space world?

- Make more similar spaceships
- More planets
- Asteroids, rocks and debris

Some Tips:

```
void Space1::draw()
{
    glPointSize(1.0);
    srand(1); /* fixed random seed, thus
    // the point are fixed
    /* comment this out to get
    // 'force field' effect
    glColor3f(1.0f, 1.0f, 1.0f);
    GLfloat x,y,z;
    glBegin(GL_POINTS);
    for (int i=0; i<1000; ++i)
    {
        x = (rand()%10000)/10000.0 * 40.0 - 20.0;
        y = (rand()%10000)/10000.0 * 40.0 - 20.0;
        z = 0;
        //try this:
        z = (rand()%10000)/10000.0 * 40.0 - 20.0;
        glVertex3f(x,y,z);
    }
    glEnd();
}
```

```
void Space1::draw1()
{
    glDisable(GL_CULL_FACE);

    //Quadric Object
    GLUquadricObj *pObj;

    //create and initialize quadric
    pObj = gluNewQuadric();
    //gluQuadricNormals(pObj, GLU_SMOOTH);
    //gluQuadricNormals(pObj, GLU_NONE);
    gluQuadricNormals(pObj, GLU_FLAT);

    glPushMatrix();
    glColor3f(0.8f, 0.8f, 0.8f);
    glTranslatef(-15.0f, 5.0f, 0.0f);
    gluSphere(pObj, 6.0f, 36, 24);
    glPopMatrix();

    glPushMatrix();
    glColor3f(1.0f, 0.0f, 0.0f);
    glTranslatef(-2.0f, 8.0f, -10.0f);
    gluSphere(pObj, 1.0f, 12, 6);
    glPopMatrix();

    gluDeleteQuadric(pObj);    //free the quadric object

    glEnable(GL_CULL_FACE);
}
```

* END OF LAB *